

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 1

Analytical Problem Solving

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

FACULTY : Dr. D.SUDHAGAR

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

Assessment Tool Weightage: 35%

Duration: 30 minutes

Marks: 25

Code of Conduct:

I **NALIPI REDDY ROHITH/192425115** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N.Rohith

QUESTIONS:05

Total Marks:25

Analytical Problem Solving

1. Learning Algorithms (Optimization Behavior)

Answer:

A **high initial learning rate** allows gradient descent to take large steps, helping it reach the region near the global minimum quickly. Gradually **decreasing the learning rate (learning rate decay)** enables smaller updates later, allowing stable convergence.

Advantages:

- Faster convergence during early training.
- Avoids slow progress caused by a small constant learning rate.

- Fine-tunes parameters near the optimum with smaller updates.

When it outperforms a constant learning rate:

- The loss function is smooth and convex.
- The initial learning rate is not excessively large.
- A proper decay schedule (step, exponential, or cosine decay) is used.

Risks:

- If the initial learning rate is too high, the algorithm may overshoot the minimum or diverge.
- If the learning rate decays too quickly, training may stop before reaching the optimal solution.

Conclusion:

A high initial learning rate with gradual decay provides both **fast learning** and **stable convergence**, often outperforming a constant learning rate when properly tuned.

2. Maximum Likelihood Estimation (MLE)

Derivation

For observations $(x_1, x_2, \dots, x_n)$ from a Gaussian distribution:

$$[X_i \sim N(\mu, \sigma^2)]$$

where (σ^2) is known.

The likelihood is:

$$[L(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)]$$

Taking the log:

$$[\begin{aligned} \log L(\mu) &= -\frac{n}{2} \log(2\pi\sigma^2) \\ &\quad - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \end{aligned}]$$

Differentiate and set to zero:

$$[\begin{aligned} \frac{d}{d\mu} \log L(\mu) &= \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = 0 \end{aligned}]$$

Hence,

$$\boxed{\hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n x_i}$$

Effect of Outliers

- The MLE is the **sample mean**.
- Outliers strongly affect the mean.
- Even one extreme value can significantly shift the estimate.

Effect of Increasing Sample Size

- As (n) increases, the estimate approaches the true mean.
- Variance decreases:

$$\boxed{\text{Var}(\hat{\mu}) = \frac{\sigma^2}{n}}$$

Thus, larger datasets produce more accurate estimates.

Robustness

MLE for Gaussian data is **not robust** because it is highly sensitive to outliers.

3. Building a Machine Learning Algorithm (Model Selection)

Challenges

- 10,000 features but only 500 samples.
- High-dimensional feature space.
- High risk of overfitting.
- Increased computational complexity.

Proposed Pipeline

1. **Data Preprocessing**
 - Normalize or standardize features.
2. **Feature Selection**
 - Remove irrelevant features using correlation filtering or L1 (Lasso).
3. **Dimensionality Reduction**
 - Apply PCA to retain the most informative components.
4. **Regularization**
 - Use L1/L2 regularization to penalize large weights.
5. **Model Training**
 - Train Logistic Regression or SVM with regularization.
6. **Cross-Validation**

- Use k-fold cross-validation for reliable evaluation.

Why it Prevents Overfitting

- Feature selection removes noise.
- PCA reduces dimensionality.
- Regularization limits model complexity.
- Cross-validation improves generalization.

Result: A simpler model with better prediction accuracy on unseen data.

4. Neural Networks (Multilayer Perceptron)

Increased Representational Power

Each hidden neuron learns a different nonlinear transformation.

More hidden neurons allow the network to model increasingly complex decision boundaries.

According to the **Universal Approximation Theorem**, an MLP with enough hidden neurons can approximate any continuous function.

Why More Neurons Do Not Always Improve Performance

- Too many neurons increase model complexity.
- The network may memorize the training data.
- This causes **overfitting**, where training accuracy is high but test accuracy is low.
- Larger networks also require more computation and training time.

Generalization

Good generalization is achieved by balancing model complexity and training data.

Regularization, dropout, and early stopping help improve generalization.

Conclusion:

More hidden neurons increase learning capacity but may reduce performance if they cause overfitting.

5. Backpropagation, SGD & Curse of Dimensionality

Impact on Gradient Updates

- In high dimensions, gradients become noisy.
- Many features contribute little useful information.
- Gradient updates become unstable and inefficient.

Impact on SGD Convergence

- More parameters increase optimization time.
- SGD requires more iterations to converge.
- Sparse data leads to slower learning.

Impact on Generalization

- High-dimensional models easily memorize training data.
- Prediction accuracy on unseen data decreases.

Techniques to Overcome These Problems

1. Dimensionality Reduction (PCA)

- Reduces redundant features.
- Speeds up gradient computation.
- Improves convergence.

2. Regularization (L1/L2, Dropout)

- Penalizes overly complex models.
- Reduces overfitting.
- Improves generalization.

Additional Techniques

- Feature Selection
- Batch Normalization
- Early Stopping
- Data Augmentation

Conclusion

The curse of dimensionality slows optimization and increases overfitting. Using dimensionality reduction and regularization enables faster convergence, more stable training, and better generalization.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies	Applies relevant	Applies basic concepts with	Fails to apply appropriate

		appropriate concepts to solve complex problems	concepts correctly with minor errors	limited accuracy	concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand